

Suppression of Velocity Pulsations with Resonant Controllers in Sensorless Vector Control of Permanent Magnet Synchronous Motors

Naohiko Sekine*, Hisao Kubota*

* Graduate School of Science and Technology, Meiji University, Kanagawa, Japan

Abstract— This paper proposes control methods to suppress periodic velocity pulsations caused by load torque in a rotary compressor during sensorless field-oriented control (FOC) of an interior permanent magnet synchronous motor (IPMSM). A primary contribution of this work is the introduction of two types of resonant controllers into the speed control system. These controllers effectively mitigate the periodic velocity pulsations caused by the fundamental and second harmonic components of the compressor's load torque. Stability analysis, where the dominant poles of the proposed sensorless control systems are determined at various speeds, demonstrates that the systems are internally stable over a wide speed range. Experimental results validate the suggested systems' effectiveness in suppressing the periodic velocity pulsations by evaluating both steady-state and transient responses.

Index Terms— Interior permanent magnet synchronous motor, periodic velocity pulsations, resonant controller, sensorless field-oriented control.

I. INTRODUCTION

Sensorless control of a permanent magnet synchronous motor (PMSM) reduces costs and improves reliability, making it widely used in various industries. Sensorless vector control achieves high efficiency and high-performance control. This method is applicable to railroads, home appliances, and more.

The load torque of a rotary compressor in an air conditioner fluctuates periodically with the rotation angle, causing periodic velocity pulsations in the motor speed. These periodic velocity pulsations lead to vibration and noise generation.

A simple solution to this problem is to introduce a sinusoidal internal model into the speed controller of the PMSM's sensorless control system [1]. The sinusoidal internal model is a sine or cosine function with infinite gain at the resonance frequency, which is used to track the AC reference signal of an active filter [2]. In the PMSM's sensorless control for compressor drives, this model, connected in parallel with the PI controller in the speed control loop, eliminates the speed deviation caused by the load torque [1].

Applying a low-pass filter (LPF) or band-pass filter

(BPF) to the speed control is also an effective strategy [3]. These filters are referred to as resonant controllers. The controllers have a quality factor, which determines the gain at the resonance frequency. Stability analysis, such as obtaining the dominant poles and stability margins, examines the internal stability of the sensorless control systems with these controllers. The authors in [3] concluded that a control system with the LPF is effective for driving over a wide speed range, while a control system with the BPF requires gain adjustment of the controller.

Another technique is to apply a combination of the BPF and equivalent derivative to the speed control [4].

There is also a method for stopping a compressor drive system with low vibration [5]. The BPF estimates the acceleration, which is required to stop the motor by this method.

Both methods in [1], [3], and [4] focus only on the fundamental component of the compressor's load torque waveform. However, experimental results in [1] and [3] revealed that the periodic velocity pulsations caused by the second harmonic component still persist.

This paper proposes control systems that also consider this second harmonic component to achieve better performance than the method in [3]. Furthermore, stability analysis, which examines the dominant poles of the systems, indicates that the proposed methods can operate over a wide speed range without requiring gain adjustment. Experimental results on steady-state and transient responses demonstrate the effectiveness of the proposed control methods.

II. SYSTEM CONFIGURATION

A. Load Torque of a Rotary Compressor

Fig. 1 shows an example of the load torque waveform of a rotary compressor. The load torque fluctuates with each rotation of the motor. Fig. 2 shows a Fourier analysis of this waveform. The waveform has a relatively large DC component, a fundamental component, and a second harmonic component. In general, the speed controller for sensorless vector control utilizes a PI compensation. The problem is that the PI compensation is unable to

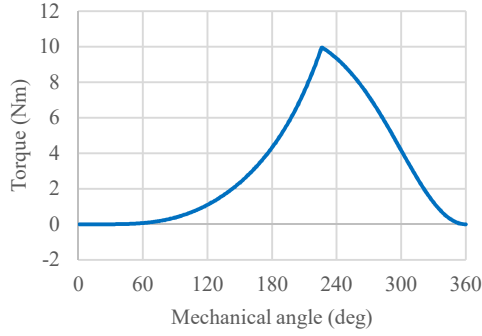


Fig. 1. Waveform of the load torque.

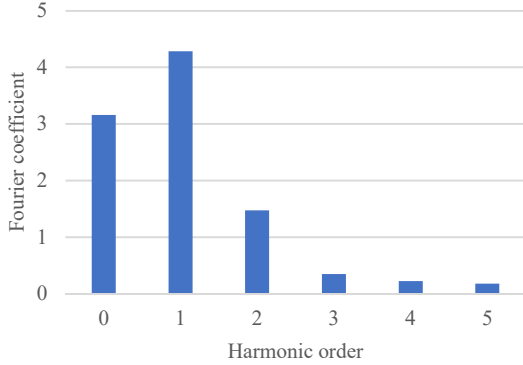


Fig. 2. Fourier analysis result of load torque waveform.

sufficiently reduce the periodic speed deviation caused by the AC components.

As shown in the analysis in Fig. 2, the 3rd and higher harmonic components of the load torque are so small that they can be ignored. The following will discuss control methods for suppressing the periodic velocity pulsations caused by the fundamental and second harmonic components of the compressor's load torque.

B. Resonant controllers

The previous research has proposed the following two types of resonant controllers [3]:

$$C_{LPF}(s) = K_{sp} \left(1 - \frac{\omega_{rc1}^2}{s^2 + \frac{1}{Q_1} \omega_{rc1} s + \omega_{rc1}^2} \right) \quad (1)$$

$$C_{BPF}(s) = -K_{sp} \frac{\omega_{rc1} s}{s^2 + \frac{1}{Q_1} \omega_{rc1} s + \omega_{rc1}^2} \quad (2)$$

where s is the Laplace operator, K_{sp} is the gain of the speed controller, ω_{rc1} is the resonance frequency, and Q_1 is the quality factor.

The sine and cosine functions in [1] and [2] are equivalent to a resonant controller with Q_1 set to infinity.

Fig. 3 shows a Bode diagram of the transfer functions in (1) and (2), normalized and with $Q_1 = 1000$. The quality factor Q_1 is the gain at the resonance frequency ω_{rc1} (rad/s). The resonance frequency ω_{rc1} is set to the reference signal of the machine angular velocity ω_m^* (rad/s). The LPF-type resonant controller in (1) is almost equivalent to a HPF by taking the form of subtracting the LPF from 1. The gain of the BPF-type resonant controller

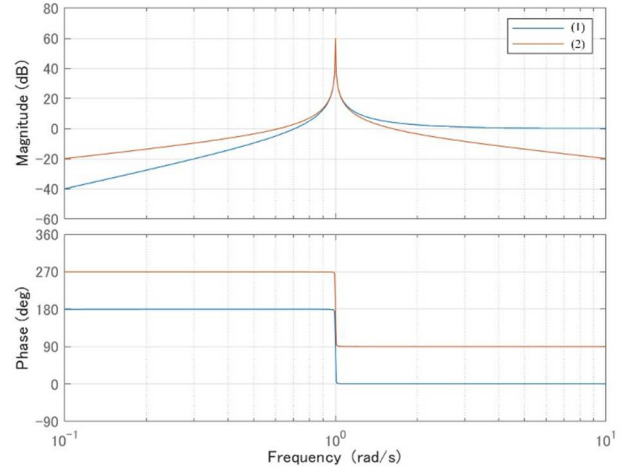


Fig. 3. Bode diagram of the transfer functions in (1) and (2) at $K_{sp} = 1$, $\omega_{rc1} = 1$, and $Q_1 = 1000$.

in (2) attenuates at both low and high frequencies.

Before considering the second harmonic component of the compressor load torque, this paper discusses another issue with the resonant controllers in (1) and (2).

The LPF-type resonant controller in (1) does not attenuate the gain in the high-frequency range. Given that the resonant controllers only need to have a significantly large gain at the resonance frequency, they should be designed in such a way that the gain is attenuated in the other frequency ranges. The control system using the BPF in (2) has a slow convergence time in transient response. Furthermore, to ensure internal stability, it is necessary to change the gain K_{sp} according to the speed reference signal or increase the gain of the PLL [3].

This paper proposes resonant controllers that address the aforementioned issues, as indicated by the following transfer functions:

$$C_{LPF1}(s) = C_{lpf1}(s)C_{filter1}(s) \quad (3)$$

$$C_{BPF1}(s) = C_{bpf1}(s)C_{filter2}(s) \quad (4)$$

where

$$C_{lpf1}(s) = -K_{sp} \frac{\omega_{rc1}^2}{s^2 + \frac{1}{Q_1} \omega_{rc1} s + \omega_{rc1}^2} \quad (5)$$

$$C_{bpf1}(s) = -K_{sp} \frac{\omega_{rc1} s}{s^2 + \frac{1}{Q_1} \omega_{rc1} s + \omega_{rc1}^2} \quad (6)$$

$$C_{filter1}(s) = 1 - \frac{\omega_{rc1}^2}{s^2 + \frac{1}{Q_2} \omega_{rc1} s + \omega_{rc1}^2} \quad (7)$$

$$C_{filter2}(s) = \frac{(2\omega_{rc1})^2}{s^2 + \frac{1}{Q_2} (2\omega_{rc1}) s + (2\omega_{rc1})^2} \quad (8)$$

The symbol Q_2 in (7) and (8) is the quality factor set to 0.5.

Fig. 4 illustrates a Bode diagram of the transfer functions in (3) and (4), normalized and with $Q_1 = 1000$. The gain of the controllers $C_{lpf1}(s)$ and $C_{bpf1}(s)$ at the resonance frequency $\omega_{rc1} = 1$ (rad/s) is Q . This characteristic serves to suppress the periodic velocity pulsations caused by the fundamental component of the load torque.

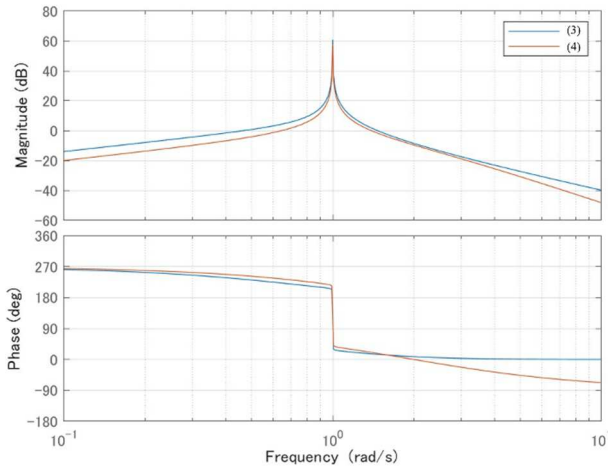


Fig. 4. Bode diagram of the transfer functions in (3) and (4) at $K_{sp} = 1$, $\omega_{rc1} = 1$, and $Q_1 = 1000$.

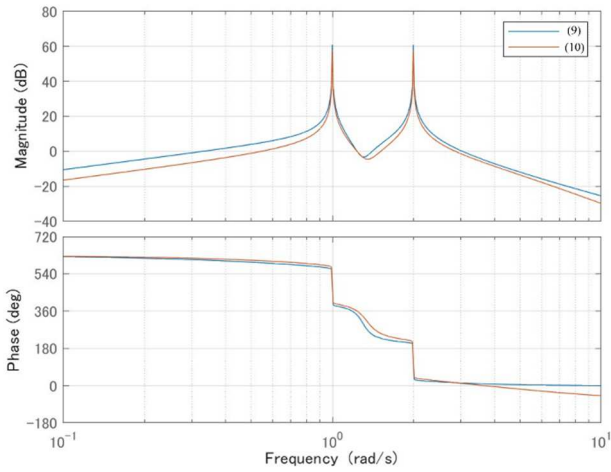


Fig. 5. Bode diagram of the transfer functions in (9) and (10) at $K_{sp} = 1$, $\omega_{rc1} = 1$, $\omega_{rc2} = 2$, and $Q_1 = 1000$.

Note that the LPF-type resonant controller in (3) has an additional filter $C_{filter1}(s)$. $C_{filter1}(s)$ in (7), which is almost equivalent to a HPF, attenuates the gain in the frequency range lower than the resonance frequency ω_{rc1} (rad/s). The gain of the controller in (3) are attenuated in bands other than the resonance frequency.

An additional filter $C_{filter2}(s)$ in the BPF-type resonant controller increases the stability margin of the system at low speed. The resonance frequency of $C_{filter2}(s)$ in (8) is set to $2\omega_{rc1}$ (rad/s) so as not to impede the performance of the controller in (6). $C_{filter2}(s)$ is a second-order delay LPF with the quality factor Q_2 of 0.5, which does not function to suppress the periodic velocity pulsations.

The experimental results in [1] and [3] showed that although the resonant controllers can suppress the periodic velocity pulsations better than the PI compensation alone, those caused by the second harmonic component of the load torque remain.

Therefore, this paper proposes resonant controllers that also suppress the second harmonic component of these periodic velocity pulsations, as shown in the following transfer functions:

$$C_{LPF2}(s) = C_{lpf1}(s)C_{filter1}(s) + C_{lpf2}(s)C_{filter3}(s) \quad (9)$$

$$C_{BPF2}(s) = C_{bpf1}(s)C_{filter2}(s) + C_{bpf2}(s)C_{filter4}(s) \quad (10)$$

where

$$C_{lpf2}(s) = -K_{sp} \frac{\omega_{rc2}^2}{s^2 + \frac{1}{Q_1} \omega_{rc2} s + \omega_{rc2}^2} \quad (11)$$

$$C_{bpf2}(s) = -K_{sp} \frac{\omega_{rc2} s}{s^2 + \frac{1}{Q_1} \omega_{rc2} s + \omega_{rc2}^2} \quad (12)$$

$$C_{filter3}(s) = 1 - \frac{\omega_{rc2}^2}{s^2 + \frac{1}{Q_2} \omega_{rc2} s + \omega_{rc2}^2} \quad (13)$$

$$C_{filter4}(s) = \frac{(2\omega_{rc2})^2}{s^2 + \frac{1}{Q_2} (2\omega_{rc2}) s + (2\omega_{rc2})^2} \quad (14)$$

The symbol ω_{rc2} in (11) and (12) is the resonance frequency.

By setting the resonance frequency ω_{rc2} (rad/s) to $2\omega_{rc1} = 2\omega_m^*$ (rad/s), the second term on the right side of (9) and (10) serves to suppress the second harmonic component of the periodic velocity pulsations.

Fig. 5 depicts a Bode diagram of the transfer functions in (9) and (10), normalized and with $Q_1 = 1000$. The gain is Q_1 at the resonance frequencies $\omega_{rc1} = 1$ (rad/s) and $\omega_{rc2} = 2$ (rad/s). This characteristic functions to suppress the periodic velocity pulsations caused by the fundamental and second harmonic component of the load torque.

C. Position and velocity estimation

The voltage equation in the $d-q$ coordinate system of the PMSM is expressed by the extended electromotive force (extended EMF) model [5], [6] as

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} R + pL_d & -\omega L_q \\ \omega L_q & R + pL_d \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} 0 \\ E_{ex} \end{bmatrix} \quad (15)$$

$$E_{ex} = (L_d - L_q)(\omega i_d - p i_q) + \omega \varphi \quad (16)$$

where v_d and v_q are the d - and q -axes stator voltage, i_d and i_q are the d - and q -axes stator current, R is the armature resistance, ω is the angular velocity at electrical angle, L_d and L_q are the d - and q -axes inductance, φ is the magnet flux linkage, and p is the differential operator.

The extended EMF model in the estimated $\gamma-\delta$ coordinate system becomes

$$\begin{bmatrix} v_\gamma \\ v_\delta \end{bmatrix} = \begin{bmatrix} R + pL_d & -\omega L_q \\ \omega L_q & R + pL_d \end{bmatrix} \begin{bmatrix} i_\gamma \\ i_\delta \end{bmatrix} + \begin{bmatrix} e_\gamma \\ e_\delta \end{bmatrix} \quad (17)$$

$$\begin{bmatrix} e_\gamma \\ e_\delta \end{bmatrix} = E_{ex} \begin{bmatrix} -\sin \Delta\theta \\ \cos \Delta\theta \end{bmatrix} \quad (18)$$

where $\Delta\theta$ is the estimated position error.

From the above equation, the estimated position error is obtained as follows:

$$\Delta\theta = \arctan\left(\frac{-e_\gamma}{e_\delta}\right). \quad (19)$$

Fig. 6 shows a block diagram of the speed and position estimator. In Fig. 6, θ (deg) is the electrical angle and $\hat{\theta}$ above the symbol indicates the estimated value. $C_{PLL}(s)$ is the phase-locked loop (PLL), which uses the PI compensation as follows:

$$C_{PLL}(s) = \omega_{pll} \left(1 + \frac{\omega_{pllpi}}{s}\right) \quad (20)$$

where ω_{pll} is the proportional gain and ω_{pllpi} is the reciprocal of the integral time constant.

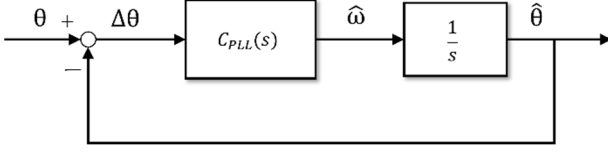


Fig. 6. Block diagram of the speed and position estimator.

The reciprocal of the integral time constant ω_{pllpi} (rad/s) is set to $\omega_{pll}/5$ (rad/s).

The transfer function from the actual speed to the estimated speed in the IPMSM is given by

$$G_{PLL}(s) = \frac{\hat{\omega}_m}{\omega_m} = \frac{\omega_{pll}s + \omega_{pll}\omega_{pllpi}}{s^2 + \omega_{pll}s + \omega_{pll}\omega_{pllpi}}. \quad (21)$$

D. Design of Current Control System

In current control, the PI compensation is performed on the d - and q -axes, respectively. The proportional gain and integral time constant of the PI controller for the d -axis component are given by

$$K_{cd} = \omega_c L_d \quad (22)$$

$$T_{cd} = \frac{L_d}{R} \quad (23)$$

where ω_c is the current control bandwidth.

The proportional gain and integral time constant of the PI controller for the q -axis component are given by

$$K_{cq} = \omega_c L_q \quad (24)$$

$$T_{cq} = \frac{L_q}{R}. \quad (25)$$

To control d - and q -axes currents independently, the decoupling control is used in this paper. In this case, the closed-loop transfer function of the current control system is expressed as

$$G_{cc}(s) = \frac{\tau}{\tau^*} = \frac{\omega_c}{s + \omega_c} \quad (26)$$

where τ is the torque of the tested IPMSM and τ^* is the torque reference signal of the tested IPMSM.

E. Configuration of Speed Control System

In speed control, the PI compensation and resonant controller are connected in parallel. The proportional gain and integral time constant of the PI controller are given by the following equations:

$$K_{sp} = J_m \omega_{sc} \quad (27)$$

$$T_{si} = \frac{1}{\omega_{spi}} \quad (28)$$

where J_m is the nominal value of the moment of inertia, ω_{sc} is the gain crossing frequency of the speed control, and ω_{spi} is the PI breakpoint angular frequency.

The proportional gain K_{sp} is also the gain of the resonant controllers in (1), (2), (3), (4), (9), and (10).

The PI breakpoint angular frequency ω_{spi} (rad/s) is set to $\omega_{sc}/5$ (rad/s).

III. STABILITY ANALYSIS

This chapter examines the internal stability of the IPMSM's sensorless control systems to which the resonant

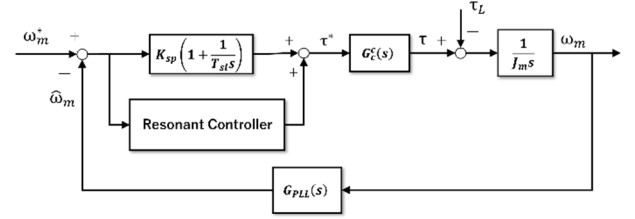


Fig. 7. Block diagram of the sensorless vector control system of the IPMSM.

 TABLE I
PARAMETERS OF THE CONTROLLERS AND ESTIMATOR

The gain crossing frequency of the speed control system ω_{sc}	10 rad/s
The PI breakpoint angular frequency ω_{spi}	2 rad/s
The quality factor Q_1	1000
The quality factor Q_2	0.5
the current control bandwidth ω_c	2000 rad/s
Proportional gain of PI controller in PLL ω_{pll}	300 rad/s
The reciprocal of the integral time constant of PI controller in PLL ω_{pllpi}	60 rad/s

 TABLE II
DOMINANT POLES OF THE SYSTEM WITH THE CONTROLLER IN (3)

		Machine angular velocity reference ω_m^* (rad/s)				
		30	60	90	150	210
J_m	$J_{mr}/3$	-1.9±j29	-1.9±j2.0	-1.8±j2.0	-1.7±j2.0	-1.7±j2.0
	$J_{mr}/2$	-3.0±j28	-3.0±j1.7	-2.8±j1.8	-2.7±j1.9	-2.6±j1.9
	J_{mr}	-2.1	-2.4	-2.5	-2.6	-2.6
	$2J_{mr}$	1.8±j18	-2.1	-2.1	-2.2	-2.2
	$3J_{mr}$	3.9±j18	-1.0±j35	-2.0	-2.1	-2.1

 TABLE III
DOMINANT POLES OF THE SYSTEM WITH THE CONTROLLER IN (4)

		Machine angular velocity reference ω_m^* (rad/s)				
		30	60	90	150	210
J_m	$J_{mr}/3$	-1.1±j29	-1.3±j59	-1.5±j89	-1.5±j150	-1.4±j210
	$J_{mr}/2$	-1.6±j28	-2.0±j59	-2.2±j89	-2.3±j150	-2.1±j210
	J_{mr}	-2.4	-2.5	-2.5	-2.7	-2.7
	$2J_{mr}$	0.56±j22	-2.2	-2.2	-2.2	-2.2
	$3J_{mr}$	2.5±j22	-2.1	-2.1	-2.1	-2.1

 TABLE IV
DOMINANT POLES OF THE SYSTEM WITH THE CONTROLLER IN (9)

		Machine angular velocity reference ω_m^* (rad/s)				
		30	60	90	150	210
J_m	$J_{mr}/3$	-2.1±j59	-2.0±j2.0	-1.9±j2.0	-1.4±j301	-1.0±j421
	$J_{mr}/2$	-2.6	-3.4±j120	-3.0±j181	-2.0±j301	-1.4±j421
	J_{mr}	-0.36±j16	-2.2	-2.4	-2.5	-2.6
	$2J_{mr}$	5.6±j17	-0.84±j31	-2.1	-2.1	-2.2
	$3J_{mr}$	8.2±j17	6.6±j32	-0.46±j45	-2.1	-2.1

 TABLE V
DOMINANT POLES OF THE SYSTEM WITH THE CONTROLLER IN (10)

		Machine angular velocity reference ω_m^* (rad/s)				
		30	60	90	150	210
J_m	$J_{mr}/3$	-1.2±j29	-1.4±j59	-1.5±j180	-1.1±j300	-0.9±j420
	$J_{mr}/2$	-1.9±j28	-2.1±j59	-2.2±j180	-1.6±j301	-1.2±j421
	J_{mr}	-2.2	-2.4	-2.5	-2.6	-2.3±j421
	$2J_{mr}$	2.5±j21	-2.1	-2.2	-2.2	-2.2
	$3J_{mr}$	4.7±j21	0.73±j42	-2.1	-2.1	-2.1

controllers are applied. Fig. 7 illustrates the block diagram of the sensorless vector control system for the IPMSM. The symbol τ_L in Fig. 7 represents the load torque. The resonant controllers vary their transfer characteristics according to the reference signal of the machine angular velocity. Therefore, it is necessary to investigate whether the control systems are internally stable for each speed reference signal.

Table I shows the controller parameters. Chapter 2 provides the designs of the controllers and estimator. The controller gains are defined in such a way that the nominal and true values of the motor parameters R , L_d , L_q , and J_m cancel each other out. Therefore, the internal stability of the control system is independent of the motor parameters as long as there are no parameter errors.

A large difference between the nominal and true values of the moment of inertia can lead to the instability of the system. Therefore, it is also important to consider the robustness of the systems against such parameter errors.

This chapter seeks to find the dominant poles of the systems. The dominant pole has the strongest influence on the system response. The fact that the real part of the dominant pole is negative indicates that the system is internally stable. The smaller the real part, the faster the response converges.

Tables II, III, IV, and V list the dominant poles of the control systems to which the transfer functions in (3), (4), (9), and (10) are applied, respectively. Here, J_{mr} is the true value of the moment of inertia. These tables provide analytical results at $J_m = J_{mr}/3$, $J_{mr}/2$, J_{mr} , $2J_{mr}$, and $3J_{mr}$ to evaluate the effect of the parameter error in the moment of inertia on the control systems. When $J_m = J_{mr}$, the systems become internally stable at $\omega_m^* \geq 30$ rad/s, and the dominant pole has the real part only. When J_m is smaller than J_{mr} , the systems are still internally stable, although there is an imaginary part at the dominant poles. As J_m is made larger than J_{mr} , the systems are no longer internally stable at low speeds.

Taking a large gain of the speed controller may cause the resonant controller to amplify the noise contained in the speed deviation, resulting in poor functionality.

In light of the above, J_m used for the gain of the speed controller should be set to a value less than or equal to the sum of the moment of inertia of the tested IPMSM and the load. This paper sets J_m to be slightly smaller than the sum of that.

IV. EXPERIMENTAL RESULTS

This chapter verifies the effectiveness of the proposed resonant controllers. Table I presents the parameters of the controllers and the estimator. Fig. 8 shows the system configuration. An induction motor (IM) on the load side, directly connected to the IPMSM, simulated the torque of the rotary compressor by torque control. Table VI lists the specifications of the tested IPMSM. The motor parameters in Table VI are the nominal values, from which the speed and current controller gains are determined. The nominal value of the moment of inertia was set to be approximately equal to the sum of the rotor moments of the tested IPMSM

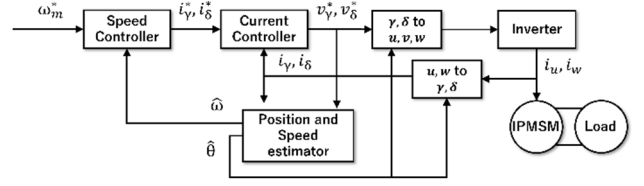


Fig. 8. System configuration of the sensorless control system for the IPMSM.

TABLE VI
SPECIFICATIONS FOR THE TEST IPMSM.

Rated power	1.5 kW
Rated current	6.3 A
Rated voltage	170 V
Rated speed	1750 min ⁻¹
Number of pole pairs	3
Stator resistance R	0.774 Ω
d -axis inductance L_d	8.90 mH
q -axis inductance L_q	12.56 mH
Magnet flux-linkage φ	0.296 Wb
Moment of inertia J_m	9.74×10^{-3} kg m ²

and the load-side IM. The reference signal of the machine angular velocity for the IPMSM was set to 90 rad/s. The d -axis current reference signal of the IPMSM was set to 0 (A). Fig. 1 plots the load torque reference signal of the IM. The carrier frequency of the inverter was set to 5 kHz. The position and speed obtained from the encoder were used for measurement. The q -axis current of the IPMSM and the load torque reference signal were measured to confirm the proper functioning of the resonant controllers.

A. Steady-state Performance

This section examines the extent to which the periodic velocity pulsations are suppressed from the actual speed.

Fig. 9 shows the steady-state performance of the control system without resonant controllers. The periodic velocity pulsations occur at about 13 rad/s. Fig. 9 indicates that the PI compensation alone fails to sufficiently suppress the periodic velocity pulsations.

Fig. 10 demonstrates the steady-state performance of the control system using the conventional resonant controller (1). The LPF-type resonant controller proposed in [3] suppresses the periodic velocity pulsations to about 3.4 rad/s. The conventional method generates some noise in the q -axis current. This is because the gain in the high-frequency range does not attenuate, as shown in Fig. 3.

Fig. 11 depicts the steady-state performance of the control systems employing the proposed resonant controllers with the resonance frequency of ω_m^* (rad/s) in (3) and (4). The system also suppresses those to about 3.5 rad/s. Figs. 11 (a) and (b) reveal that these resonant controllers are unlikely to generate noise in the q -axis current compared to the one in (1).

The waveforms of the q -axis current in Figs. 10 and 11 are sinusoidal. This indicates that these resonant controllers properly suppress the periodic velocity pulsations caused by the fundamental component of the load torque. However, Figs. 10 and 11 also proves that the periodic velocity pulsations caused by the second

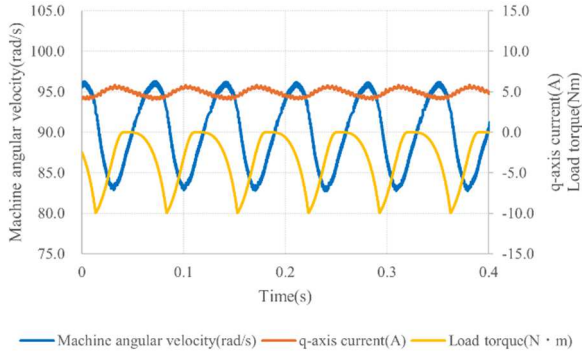


Fig. 9. Steady-state performance of the sensorless control system without resonant controllers.

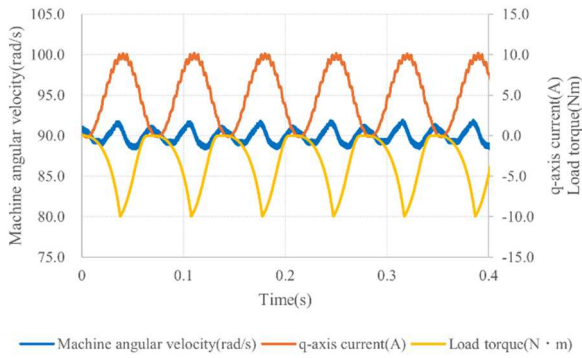


Fig. 10. Steady-state performance of the system using the conventional resonant controller in (1).

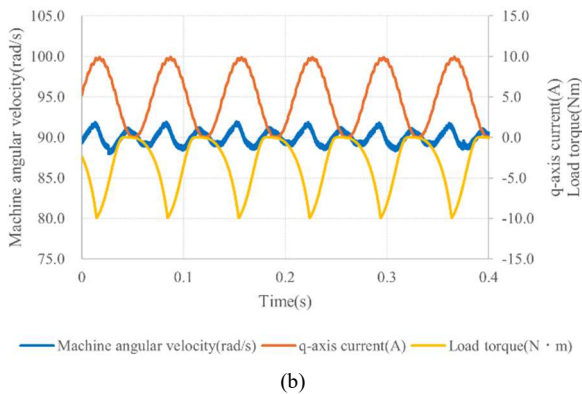
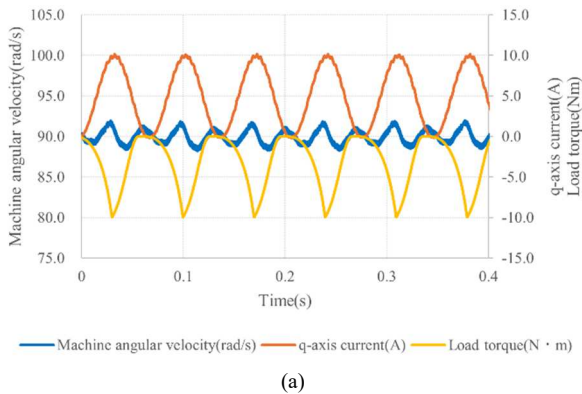
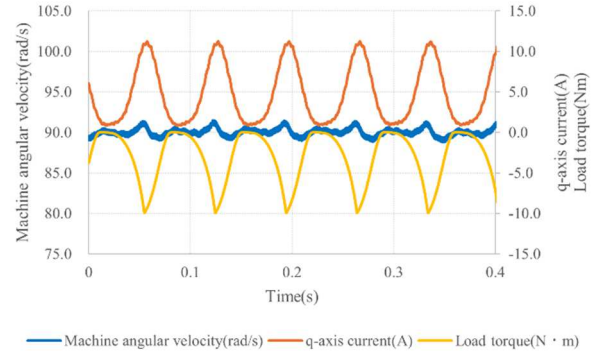
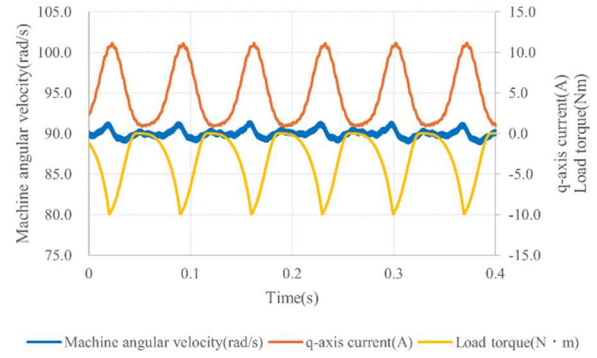


Fig. 11. Steady-state performance of the control systems using the proposed resonant controllers with the resonance frequency of ω_m^* (rad/s). (a) With the controller in (3). (b) With the controller in (4).



(a)



(b)

Fig. 12. Steady-state performance of the control systems using the proposed resonant controllers with resonance frequencies of ω_m^* and $2\omega_m^*$ (rad/s). (a) With the controller in (9). (b) With the controller in (10).

harmonic component of the load torque persists.

The proposed systems using the resonant controllers with the resonance frequencies of ω_m^* and $2\omega_m^*$ (rad/s) in (9) and (10) suppresses the periodic velocity pulsations to about 2.4 and 2.5 rad/s, respectively, as shown in Fig. 12 (a) and (b). By comparing Figs. 10, 11, and 12, it can be observed that the waveform of the q-axis current in Fig. 12 more closely matches the waveform of the load torque. This implies that the controllers (9) and (10) can suppress the second harmonic component of the periodic velocity pulsations.

B. Transient Performance

This section describes how the periodic velocity pulsations converge after the resonant controllers are turned on. The resonant controllers were activated at the time of 0.2 seconds in the figures.

Fig. 13 illustrates the transient response of the systems with the conventional resonant controller (1). The periodic velocity pulsations converged in about 0.6 seconds.

Fig. 14 plots the transient response of the systems employing the proposed resonant controllers with the resonance frequency of ω_m^* (rad/s) in (3) and (4). Fig. 15 displays the transient response of the system using the proposed resonant controllers with the resonance frequencies of ω_m^* and $2\omega_m^*$ (rad/s) in (9) and (10). In the proposed systems, the periodic velocity pulsations also converge within about 0.6 seconds.

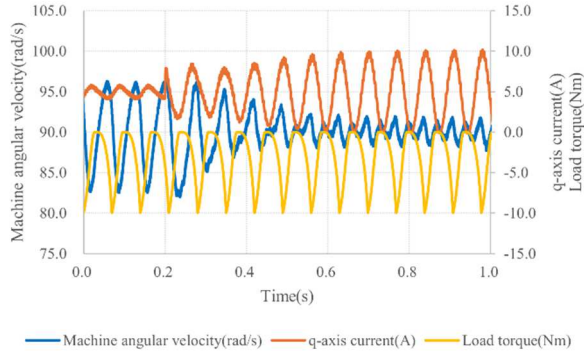
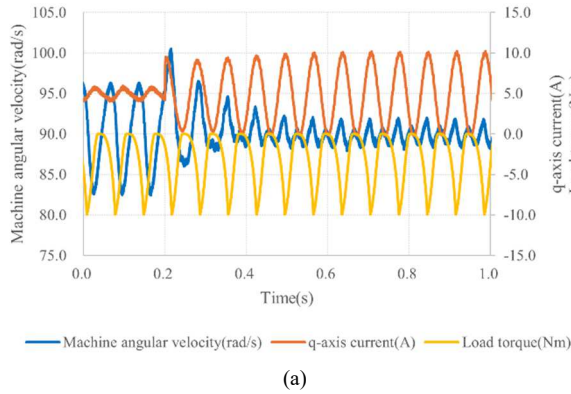
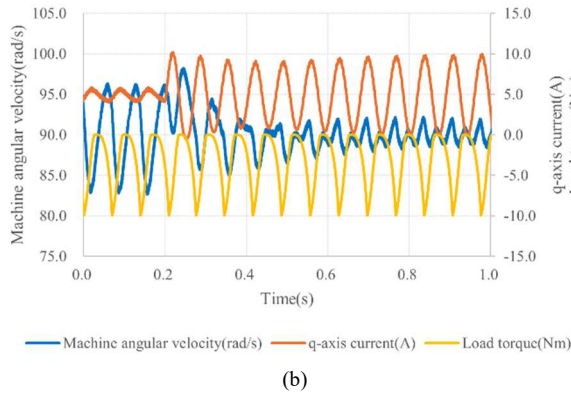


Fig. 13. Transient response of the systems with the conventional resonant controller in (3) at 90 rad/s.



(a)



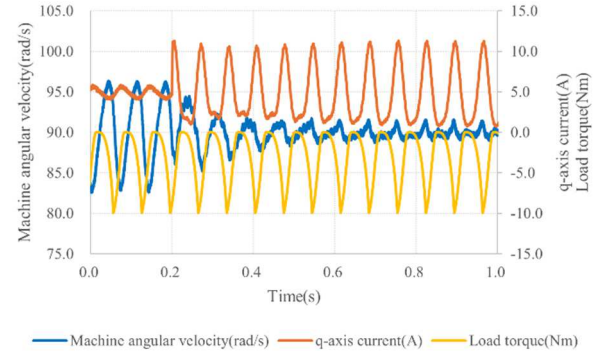
(b)

Fig. 14. Transient response of the systems with the proposed resonant controllers whose resonance frequency is ω_m^* (rad/s). (a) With the controller in (3). (b) With the controller in (4).

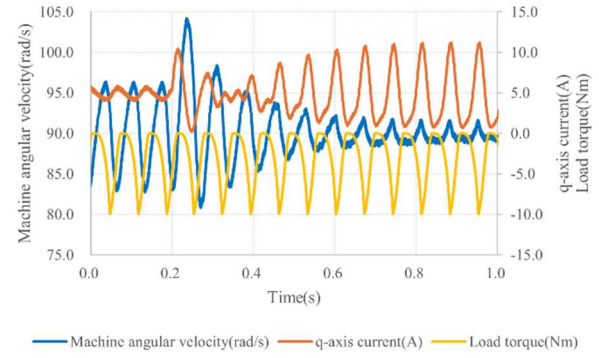
The activated resonant controllers adjust the q -axis current to suppress the periodic velocity pulsations. These experimental results prove that the systems with the proposed resonant controllers can converge the periodic velocity pulsations in almost the same time as the conventional systems.

V. CONCLUSIONS

This paper proposed the application of the resonant controllers to the sensorless vector control system for the IPMSM to suppress the periodic velocity pulsations caused by the load torque of a rotary compressor. The proposed controllers effectively mitigate the periodic velocity pulsations caused by both the fundamental and



(a)



(b)

Fig. 15. Transient response of the systems with the resonant controllers whose resonance frequencies are ω_m^* and $2\omega_m^*$ (rad/s). (a) With the controller in (9). (b) With the controller in (10).

second harmonic components of the compressor's load torque. Stability analysis demonstrated that the proposed systems remain internally stable over a wide speed range without the need for gain adjustments. The experimental results confirmed that the proposed resonant controllers can effectively suppress periodic velocity pulsations and reduce noise generation in the q -axis current compared to the conventional one.

REFERENCES

- [1] Y. Notohara, D. Li, Y. Iwaji, M. Tamura, and K. Tsukii, "Study on vibration suppression control for rotary compressor," *IEEJ Journal of Industry Applications*, vol.10, no.4, pp.481-486, 2021.
- [2] S. Fukuda and T. Yoda, "A novel current-tracking method for active filters based on a sinusoidal internal model," *IEEE Trans. Ind. Applicant.*, vol.37, no.3, pp. 888-895, 2001.
- [3] H. Kubota, M. Nishioka, N. Sekine, T. Shibayama, and T. Aizawa, "Suppression of Periodic Velocity Pulsation in Sensorless Vector Control of PMSM," *IEEJ Transactions on Industry Applications*, vol.144, No.4, pp.175-181, 2024. (in Japanese)
- [4] T. Fukumoto, H. Hamane, and Y. Hayashi, "Reduction of cyclic speed change using a bandpass filter for compressor PMSM sine-wave position sensorless drives," *Electrical Engineering in Japan*, vol.169, no.4, pp.75-84, 2009.
- [5] T. Shioi, T. Kumagai, K. Kusaka, and J. Itoh, "Low Vibration Stopping Method for the Compressor Driving Motor," *IEEJ Journal on Industry Applications*, vol.11, No.5, pp.722-729, 2022.

- [6] Z. Chen, M. Tomita, S. Doki, S. Okuma, "An extended electromotive force model for sensorless control of interior permanent-magnet synchronous motors," *IEEE Trans. Ind. Electron.*, vol.50, no.2, pp. 288-295, 2003.
- [7] S. Morimoto, K. Kawamoto, M. Sanada, and Y. Takeda, "Sensorless control strategy for salient-pole PMSM based on extended EMF in rotating reference frame," *IEEE Trans. Ind. Appl.*, vol.38, no.4, pp. 1054-1061, 2002.